Enumerative Combinatorics — problem 5.22

A. Engel — Problem-Solving Strategies · 1998

Find combinatorial proofs of the following formulas. Use bijection or counting the same objects in two ways.

(a) $\binom{n}{s} = \frac{n}{s}\binom{n-1}{s-1}$

(b) $\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$

(c) $\sum_{i=0}^{n}\binom{n}{i}\binom{n}{n-i} = \binom{2n}{n} = \sum_{i=0}^{n}\binom{n}{i}^2$

(d) $\binom{n}{s} = \frac{n}{n-s}\binom{n-1}{s}$ (e) $\binom{n}{0} + \binom{n}{2} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots$

(f) $\binom{n}{0} + \binom{n+1}{1} + \binom{n+2}{2} + \cdots + \binom{n+r}{r} = \binom{n+r+1}{r}$.